AI-101 MINI PROJCT

Traffic Prediction and Congestion Alert System

Team no. 56-

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Repo link-https://github.com/2026EEB1336/Traffic-Prediction-.git

Dataset Source: Kaggle

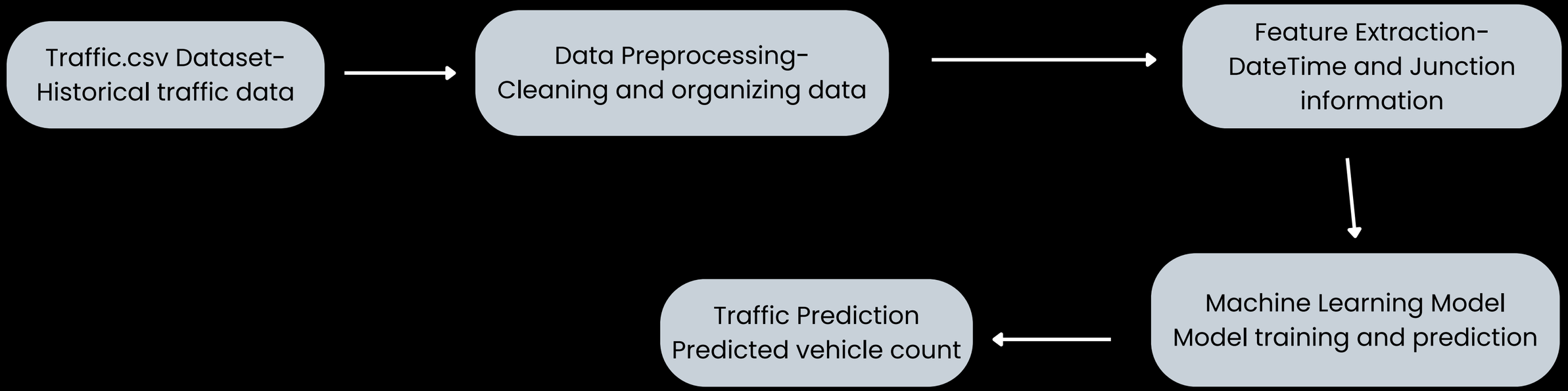
Dataset File: Traffic.csv

Dataset Type: Time-series traffic data

Dataset Columns:

- DateTime – Date and time of traffic recording
- Junction – Junction identification number
- Vehicles – Number of vehicles recorded
- ID – Unique record identification number

Problem Statement: To analyze historical traffic data and predict the number of vehicles at a traffic junction.



Libraries Used:

PANDAS

Data Handeling and Analytics

NUMPY

Numerical Calculations

MATPLOTLYB

Graphs and visualisation

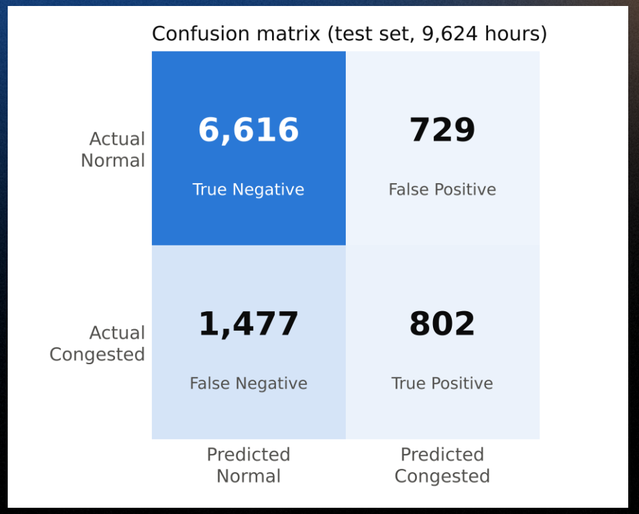
SCIKIT-LEARN

Machine Learning

Results and Output

Metric	Value
Test set size	**9,624 hours (80/20 stratified split)**
Accuracy	**77.1%**
Baseline accuracy (always Normal)	**76.3%**
Precision	**52.4%**
Recall	**35.2%**
F1-score	**0.42**
True Negative / False Positive	**6,616 / 729**
False Negative / True Positive	**1,477 / 802**
HIGH alerts (≥ 70%)	**0 of 48,120 hours (max 69.0%)**
MODERATE alerts (≥ 40%)	**12,564 hours (26.1%)**

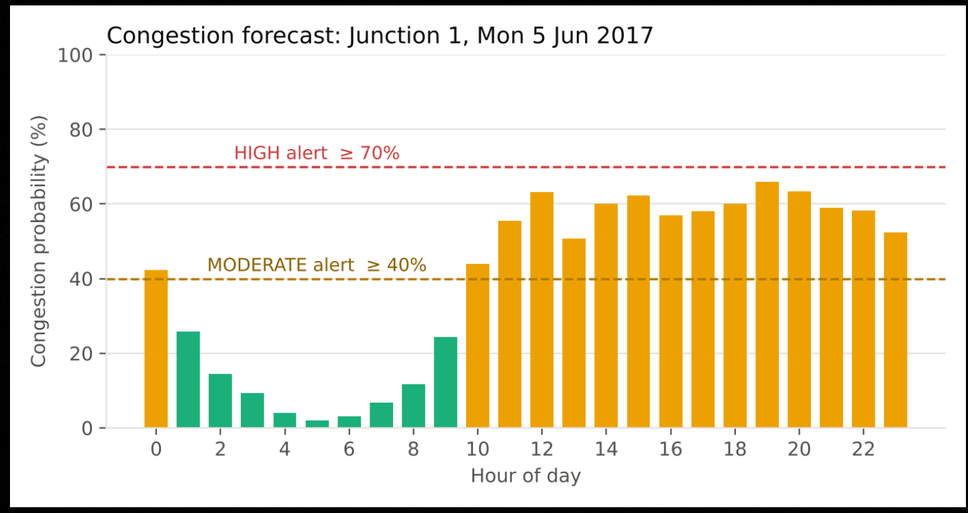
Key Metrics



Confusion Matrix

```
$ python alert_system.py
Congestion Alert System initialized successfully!
------------------------------------------------
     TRAFFIC CONGESTION ALERT
------------------------------------------------
Junction     : 1
Date & Time  : 05-06-2017 18:00
Congestion   : 60.04%
Alert Status : ● MODERATE
Vehicle Limit : 59 vehicles/hour
------------------------------------------------
------------------------------------------------
     TRAFFIC CONGESTION ALERT
------------------------------------------------
Junction     : 4
Date & Time  : 05-06-2017 18:00
Congestion   : 48.47%
Alert Status : ● MODERATE
Vehicle Limit : 9 vehicles/hour
------------------------------------------------
```

Alert System Output



Congestion Forecast

Learnings

What worked

- Time-of-day pattern captured: nights lowest, afternoon and evening highest
- Per-junction 75th-percentile label makes 'congested' comparable across junctions
- Alert output is simple: LOW, MODERATE or HIGH plus the vehicle limit

What didn't

- Accuracy 77.1% barely beats the 76.3% baseline; recall only 35%
- No trend feature: Junction 1 grew from ~20 to ~73 vehicles/hour
- HIGH alert never fires
- Next: add trend and previous-hour features

Who did what

- 2026EEB1336: repo, data loading, EDA, training script
- 2026AIB1078: congestion label, scaling, model training
- 2026MCB1564: evaluation, feature-impact plot
- 2026EEB1383: alert system